

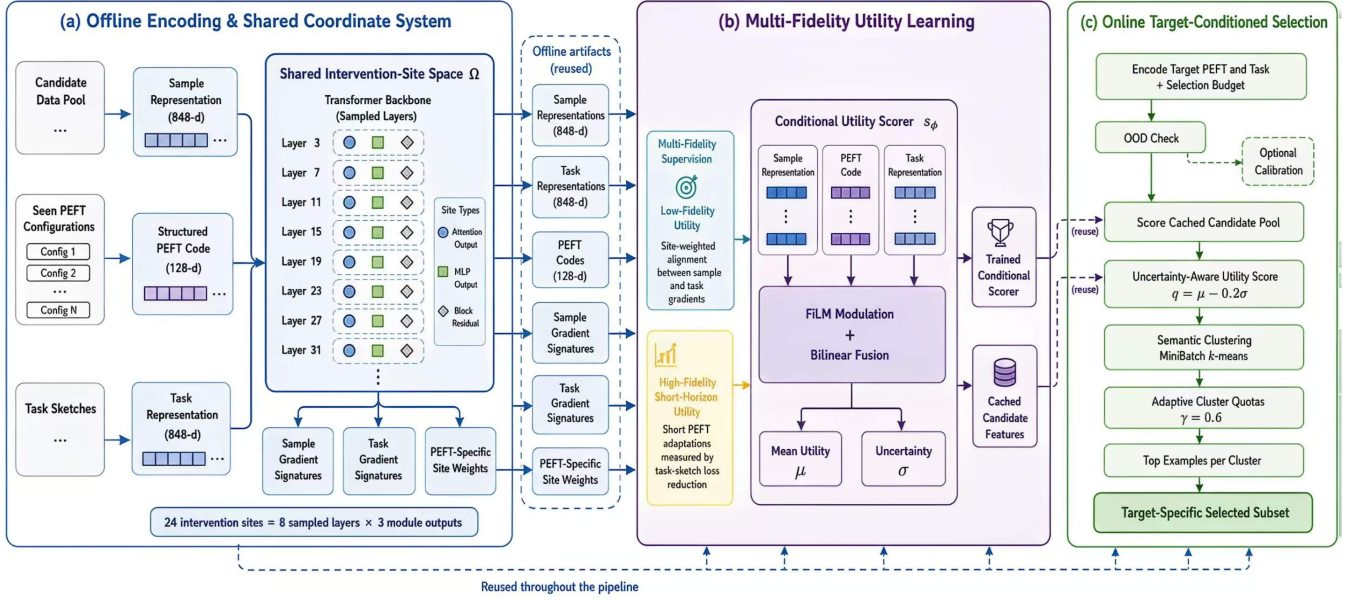


Figure 2: PCU-Select separates reusable offline supervision from online PEFT–task scoring. Its shared coordinate system Ω contains 24 intervention sites, or selected module outputs. Sample and task-sketch gradients use these site indices, while the PEFT code identifies the sites affected by each configuration. The 128-dimensional PEFT code comprises a 96-dimensional site/operator mask, 16 capacity dimensions, and 16 recipe dimensions. Low-fidelity gradient proxies provide broad coverage; high-fidelity short updates provide direct, costlier supervision.

Prefix, P-Tuning, and BitFit use the same deterministic encoding rules as seen families. The scorer has no learned family lookup or operator centroid. The support policy below decides whether a target uses direct scoring, calibrated scoring, or a per-target selector.

Training these representations requires labels for many (x, p, t) triples, but measuring every PEFT update would defeat amortization. PCU-Select instead uses *multi-fidelity supervision*, which combines labels with different costs and directness. A low-fidelity gradient proxy cheaply covers the pool. High-fidelity short updates directly measure sketch-loss reduction for fewer triples. Together, they learn broad ranking structure and correct errors near the subset cutoff.

The *gradient-proxy label* measures alignment between a sample and the task sketch at sites used by the target PEFT:

$$u^{\text{lo}}(x, p, t) = \sum_{\omega \in \Omega} \tilde{w}_p^\omega \cos(g_x^\omega, g_t^\omega). \quad (6)$$

Here g_x^ω is the response-token loss gradient at site ω . A Johnson–Lindenstrauss random projection (Johnson and Lindenstrauss 1984) compresses it to 256 dimensions and unit-normalizes the result. The task-sketch mean gives g_t^ω . Equation (6) therefore measures alignment only at sites used by p . All sketches use response-token cross-entropy for these signatures; task-native metrics such as option likelihood, Pass@1, and F1 remain downstream evaluation metrics. Because cosine values can be negative, PCU-Select rank-normalizes them within each (p, t) bucket before proxy distillation and ranking. Cached sample signatures then label many (x, p, t) triples without new gradient extraction.

The *short-update label* measures realized sketch-loss reduction near the subset cutoff. For each sampled triple, PCU-Select uses two anchors θ_a : the base model and a warm checkpoint after adaptation has begun. The horizon h is the number of short-update steps. A mini-batch $\mathcal{B}(x)$ contains the sampled example and seven condition-matched fillers. PCU-Select applies PEFT p for h steps and measures

$$\Delta(x, p, t, a, h) = L_{V_t}(\theta_a) - L_{V_t}(\text{Adapt}_p^h(\mathcal{B}(x); \theta_a)). \quad (7)$$

Thus, Δ is the task-sketch loss before the update minus its loss afterward. Rank-normalized reductions for $h \in \{1, 4\}$ from both anchors define u^{hi} . The fillers make this mini-batch effect more stable than a single-example update. PCU-Select allocates 10K short-update labels in three stages. Coverage sampling first spreads labels across the registry without a scorer.

Coverage labels and the proxy then train a provisional scorer. It predicts ranks and a per-triple uncertainty scale, an estimate of prediction noise. Uncertainty sampling selects triples with large scales. Boundary sampling selects large predicted–observed rank disagreements near the cutoff. Every rank normalization compares labels within one PEFT, task, anchor, and horizon bucket.

The proxy cheaply orders the full pool, while short updates target consequential ranking errors. Two anchors test whether utility persists after adaptation begins; two horizons reduce dependence on one update length. Ranking within condition-matched buckets avoids assuming a shared raw-loss scale across tasks or configurations.